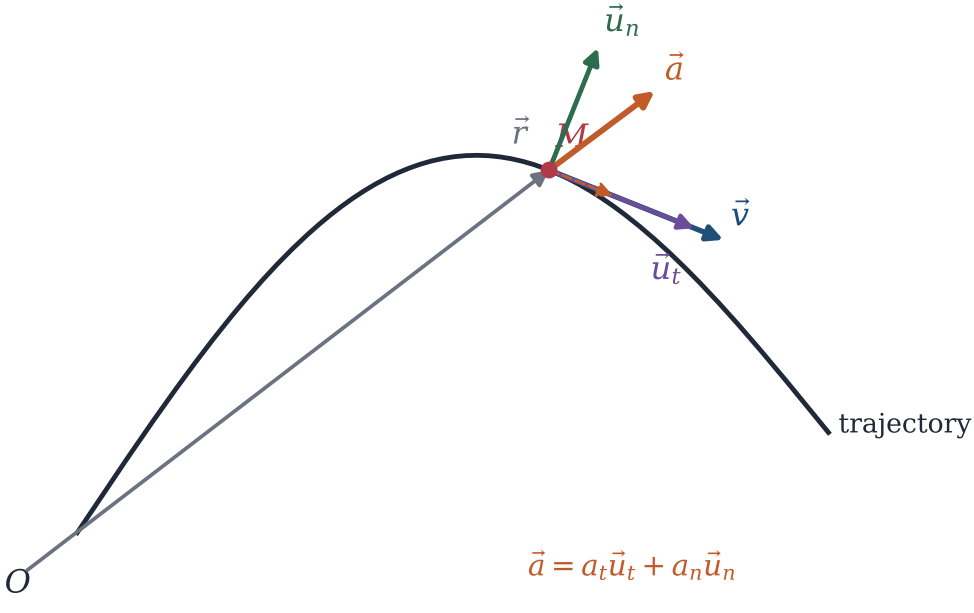

$$\vec{a} = a_t\vec{u}_t + a_n\vec{u}_n$$